مدينة زويل للعلوم والتكنولوجيا
Zewail City of Science and Technology

Computer-Aided Design project (SPC 470)

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Wind Turbine Blade Design

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1. Problem Description

This report presents a comprehensive numerical investigation of a Horizontal Axis Wind Turbine (HAWT) blade developed in a previous assignment using the **S834** airfoil. The blade geometry was optimized using QBlade and modeled in SolidWorks. The study evaluates aerodynamic performance, structural response, and dynamic behavior under specified operating conditions. A three-dimensional steady CFD analysis is performed at a wind speed of 9 m/s and a rotational speed of 400 rpm to obtain flow fields and aerodynamic loads. A grid independent study is conducted to ensure numerical accuracy. The resulting pressure loads are transferred to a structural model assuming wood material properties through a one-way fluid–structure interaction (FSI) approach to compute stresses and deformations. A modal analysis is then performed to determine natural frequencies and mode shapes, and potential resonance risks are assessed by comparing modal frequencies with operational excitations.

2. Geometry and Blade Design Basis

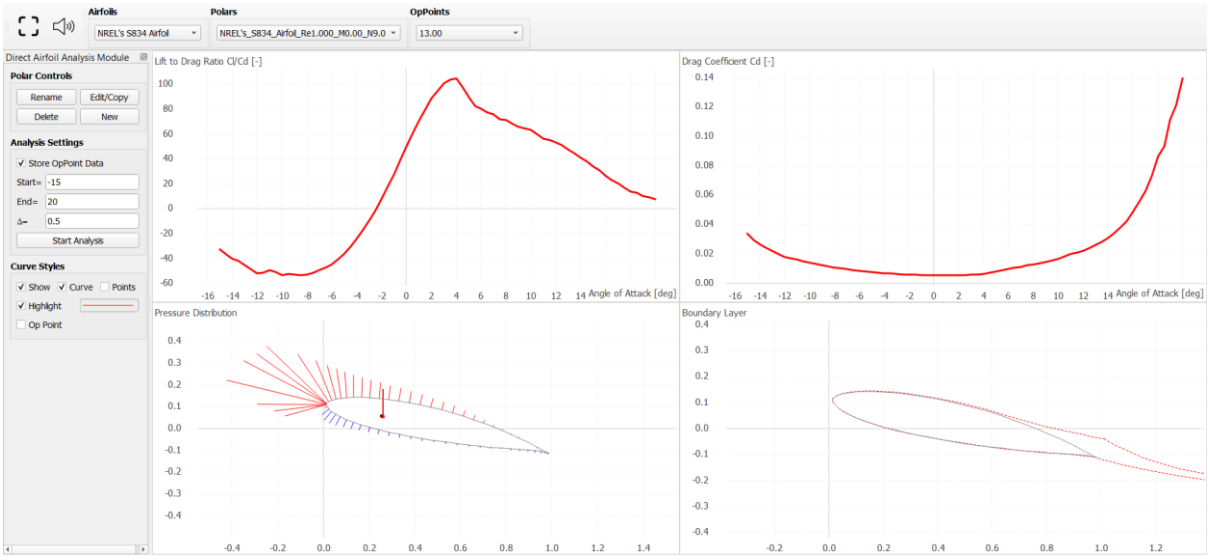
CAD Design Process:

1- Airfoil Selection: S834

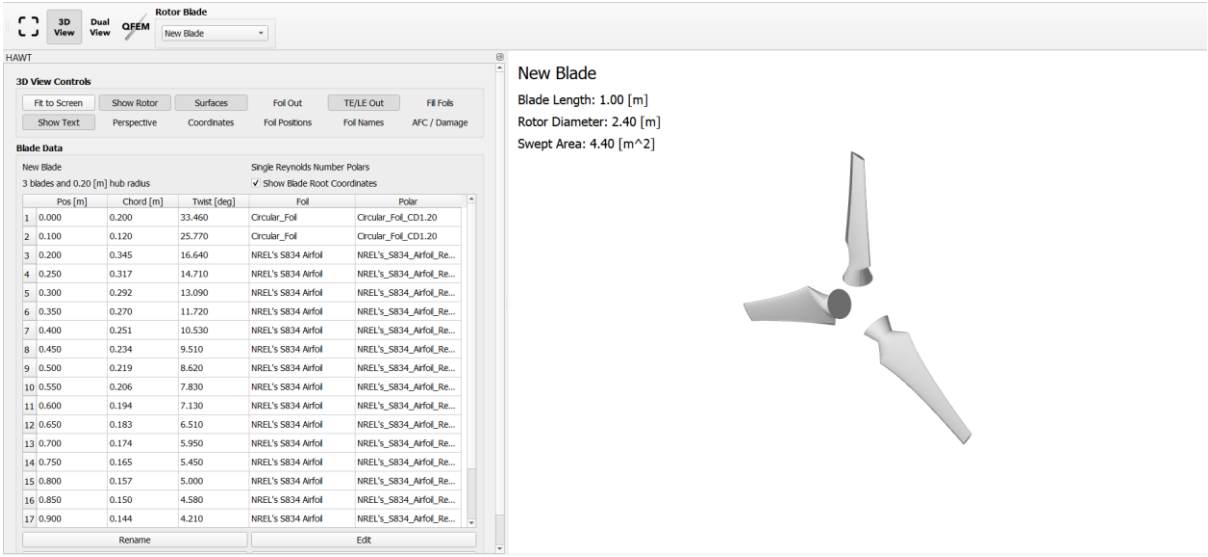
2- Using *QBlade* Software to determine the chord and twist angle at each section:

   a. Downloading S834 x-y coordinates from *Airfoil Tools*

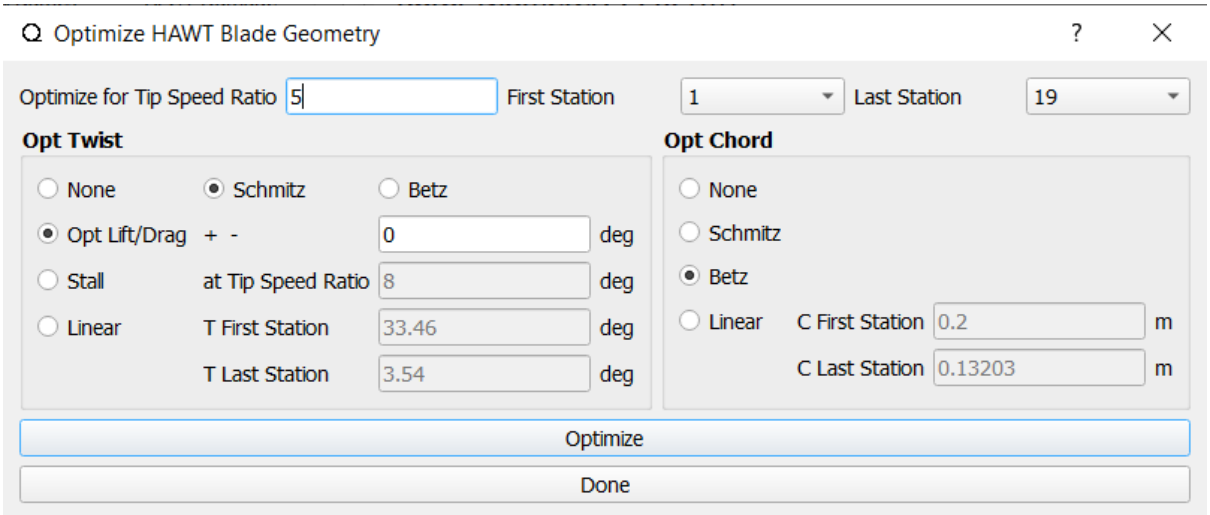
   b. Importing the coordinates in *QBlade* and perform an analysis on the airfoil



c. Define a new blade design and insert 19 stations (18 sections) with a 0.05-meter distance between consecutive stations.
d. Select a circular airfoil for the first two stations and S834 for the following stations.

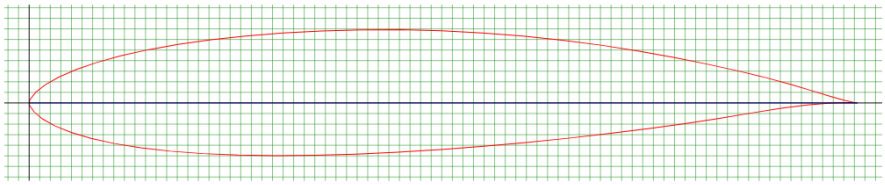


e. Optimize the chord and twist using:
    1) tip speed ratio = 5
    2) select Schmitz for twist optimization
    3) select Betz for chord optimization



3- Use Airfoil Tools plotter (http://airfoiltools.com/plotter/index) for every station with the given chord length and twist.
    a. Notes:
        1) Chord length should be converted to millimeters

2) The twist for each station should be reduced by the twist of the last station so that the tip twist is reset to zero.



Name = NREL's S834 Airfoil
Chord = 782.78mm Radius = 0mm Thickness = 100% Origin = 0% Pitch = 0°

Open full size plan in new window | Open paginated plan in new window for multi page printing
CSV file of coordinates | SVG image as text file
Restore defaults | Download PDF file

Airfoil	s834-nr - NREL's S834 Airfoil	Choose from database list or add you own airfoils here.
Chord (mm)	782.78	Chord width in millimetres. (1 inch = 25.40mm)
Radius (mm)	0	Radius of camber in millimetres, Zero for no curve
Thickness (%)	100	Thickness adjustment.100% is normal thickness. 50% is half. 200% is double
Origin (%)	0	Adjust the position of the origin e.g. 50% is mid chord
Pitch (degrees)	0	Pitch or angle of attack. 180 flips the plot
Halo (mm)	0	Line parallel to airfoil for wing covering or jig. Negative values are external, positive internal.
Halo (mm)	0	Second line parallel to airfoil as above
Colour	Colour	Colour palette or black & white
Line thickness (%)	100	Scale the line thickness (10% to 500%)
Reverse	☐	Plot a mirror image
Data box	☑	Print the airfoil data on the image
Camber line	☐	Show camber line on image

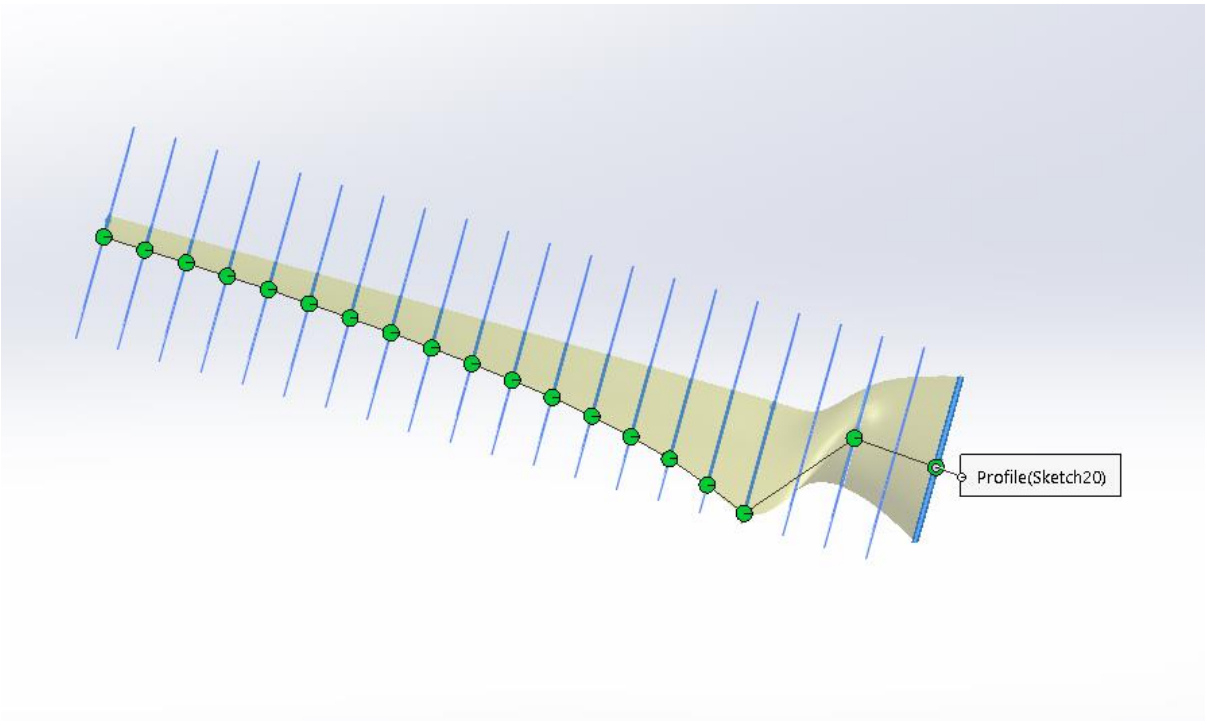
b. For each station, the CSV file is downloaded and reformatted to include the z-location (pos[m] in QBlade)

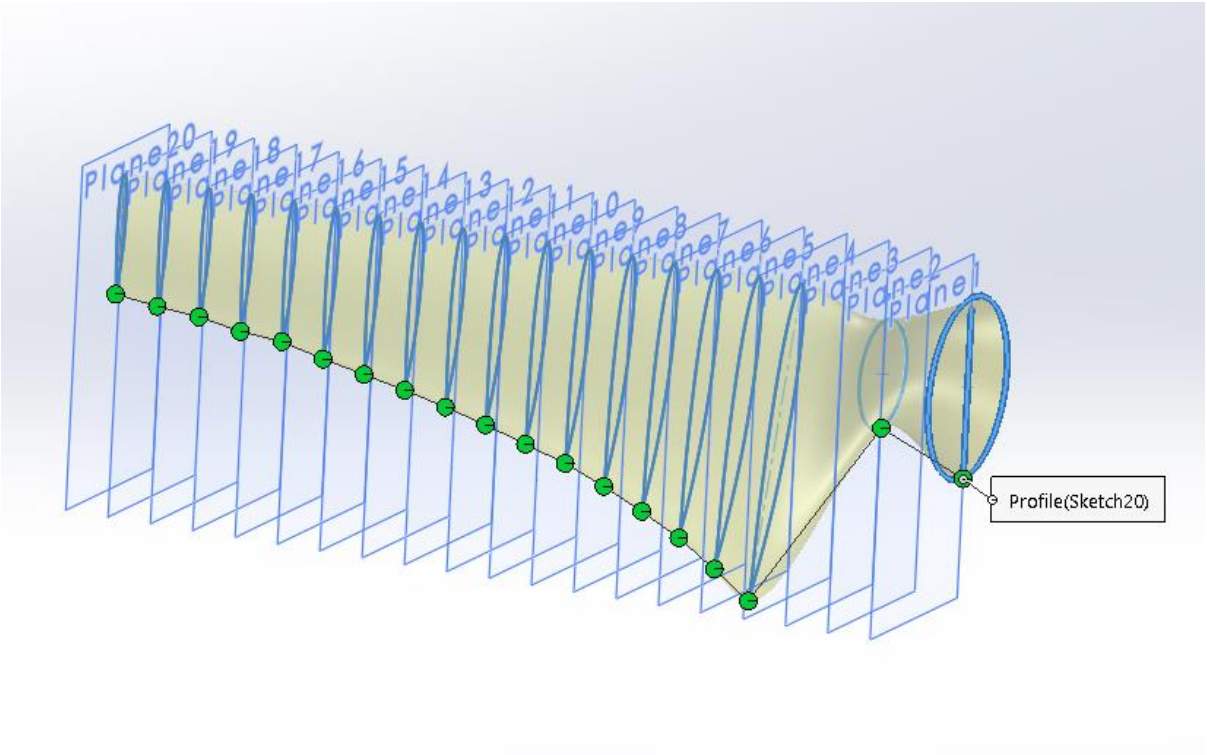
	A	B	C
1	264.2242	-54.8144	350
2	263.2174	-54.4385	350
3	260.3822	-53.1627	350
4	256.081	-50.944	350
5	250.6224	-47.9555	350
6	244.1612	-44.4596	350
7	236.7028	-40.6825	350
8	228.2052	-36.6768	350
9	218.785	-32.4552	350
10	208.5738	-28.1035	350
11	197.6878	-23.7238	350
12	186.226	-19.4172	350
13	174.276	-15.2705	350
14	161.9185	-11.3479	350
15	149.2513	-7.66643	350
16	136.4363	-4.28859	350
17	123.5509	-1.28474	350
18	110.6807	1.346102	350
19	97.96386	3.613853	350
20	85.54286	5.50606	350
21	73.55121	7.010178	350
22	62.11994	8.112248	350
23	51.36316	8.805687	350

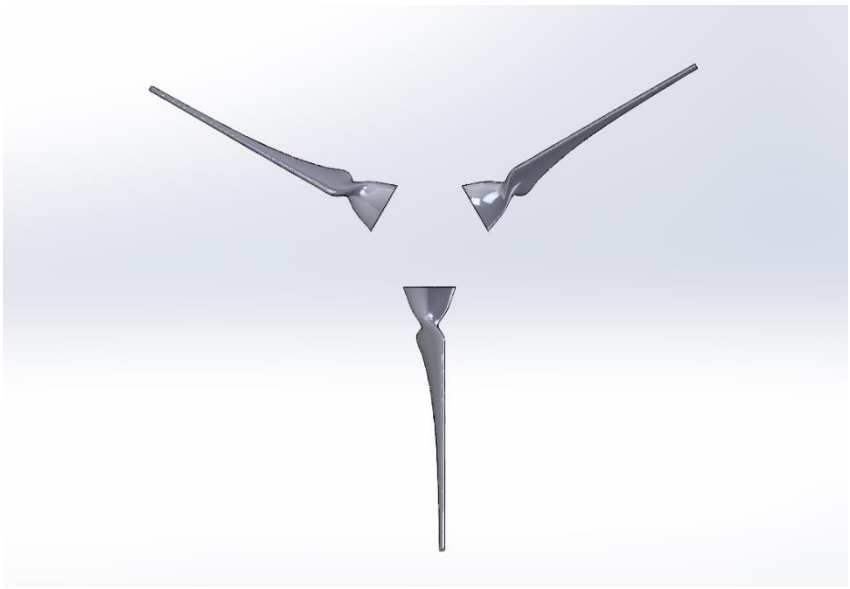
c. Save the .csv files as .txt files

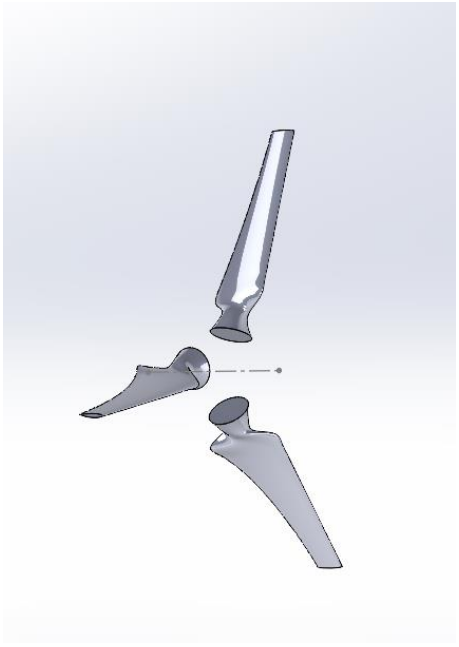
4- Insert the .txt files in *SolidWorks* and create a loft

Final Design:











3. Numerical Methodology

3.1 Processing Stages Overview

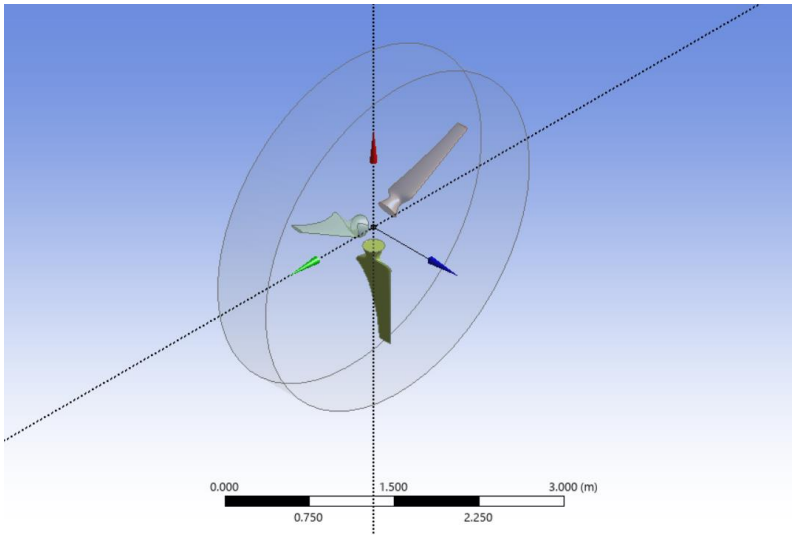
These three main stages:

- Pre-processing: Geometry import, domain creation, meshing, material definition, and boundary conditions.
- Processing: Solution of governing equations in Fluent and Structural solvers until convergence.
- Postprocessing: Extraction and analysis of contours, forces, coefficients, stresses, deformations, and modal data.

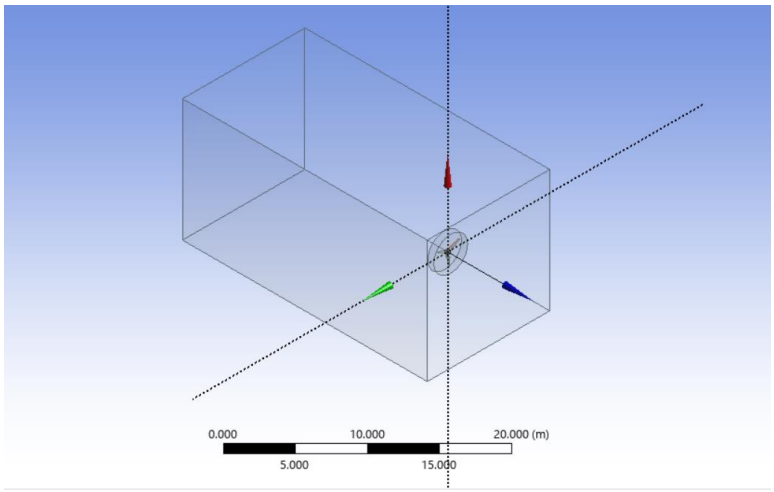
3.2 Pre-Processing

3.2.1 Computational Domain

- Circular region with dimensions:
  - Diameter: 3.3m
  - Thickness: 0.6m (symmetry)



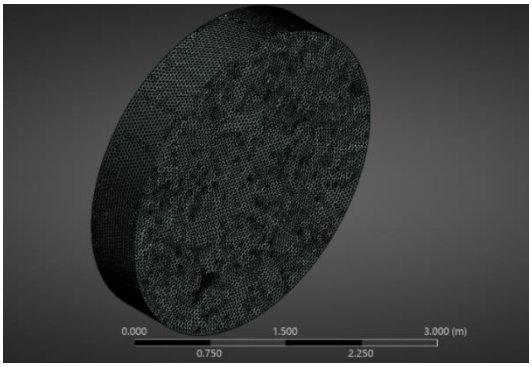
- Flow region with dimensions:
  - Frontal section: 12*12 m^2
  - Thickness ahead of the blades: 4m
  - Thickness behind the blades: 20m

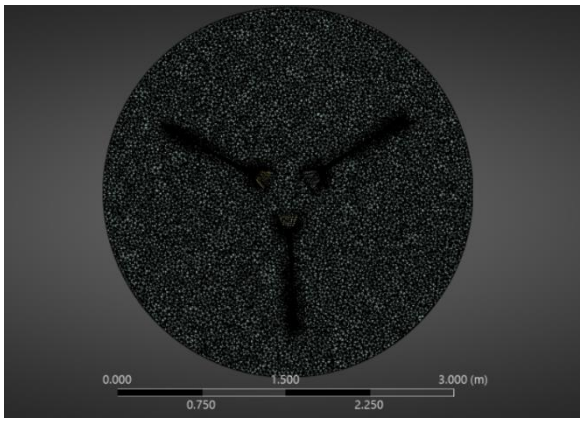


- **Comment:** the flow region size was acceptable as the flow velocity was constant at the boundaries.

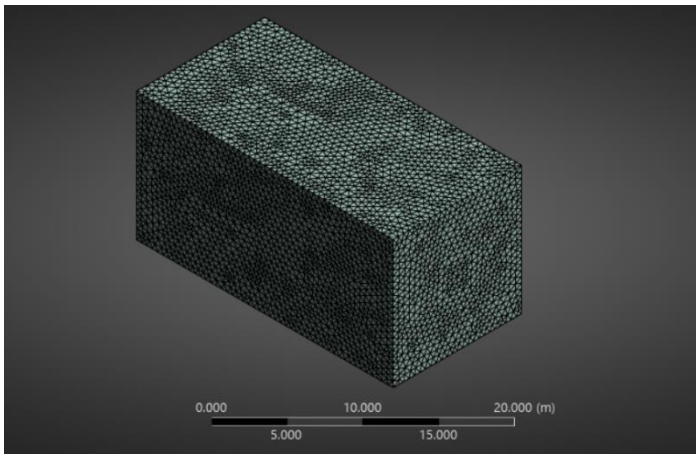
3.2.2 Mesh Generation

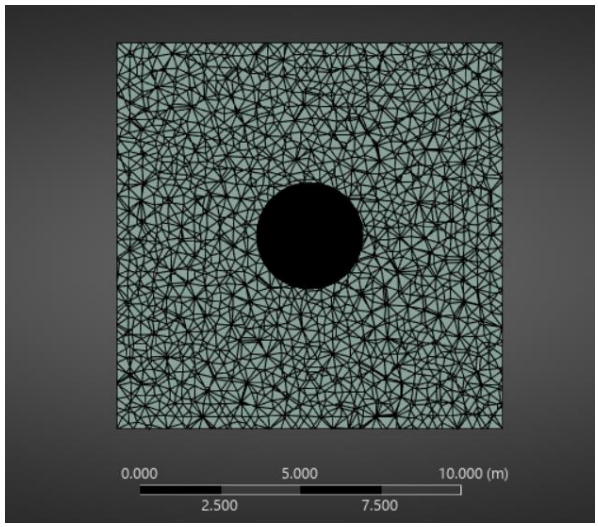
- Circular region with dimensions:
  - Mesh type: Prime Mesh
  - Element size: 0.04m





- Flow region with dimensions:
  - Mesh type: Prime Mesh
  - Element size: 0.6m

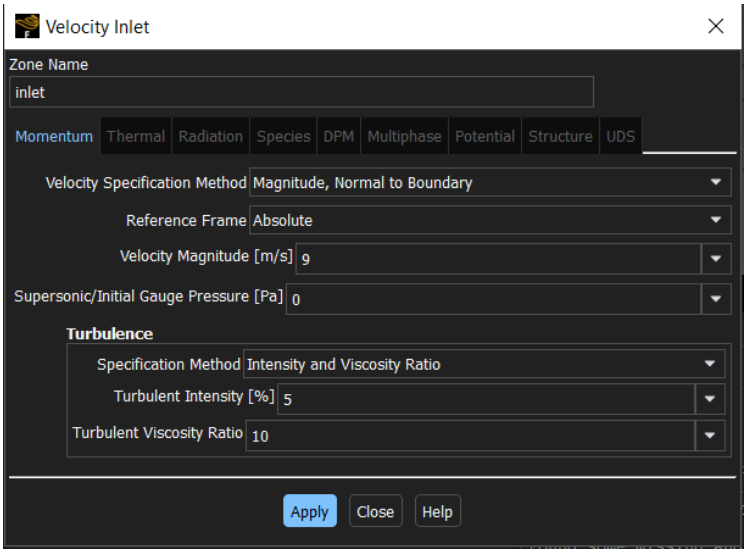




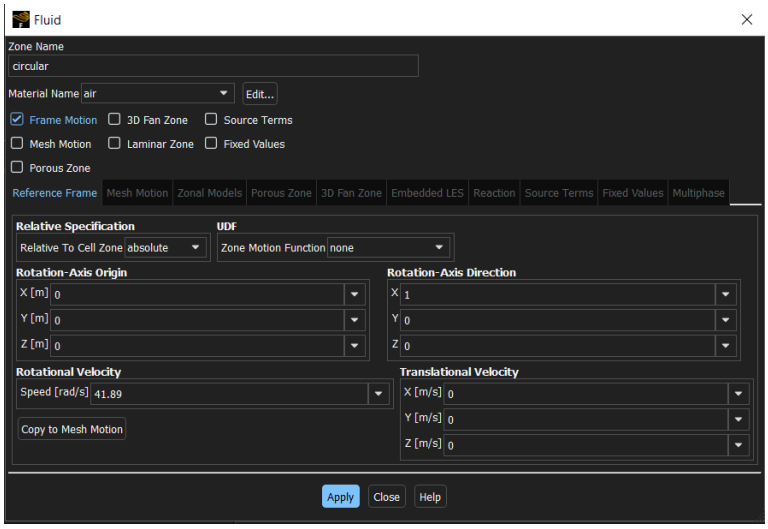
- Number of Cells: 957,346
- Comments:
  - Prime mesh type is not the best type for this simulation; however, due to the Ansys Student Version limitation, it was the only type able to create the mesh.
  - The y^+ analysis around the blades could not be done using inflation. The inflation failed because it needed specific type of meshes around it, and changing the mesh type was infeasible because of the Student Version limitations.
  - The number of cells nearly reached the Student Version limit of 1 million. In addition, changing the mesh type while keeping the element size would create a greater number of cells that would subsequently exceed the limit.

3.3 Boundary Conditions and Models

- Inlet: Velocity inlet = 9 m/s



- Outlet: Pressure outlet (0 gauge)
- Blade: No-slip wall
- Rotation: 400 rpm (41.89 rad/s)



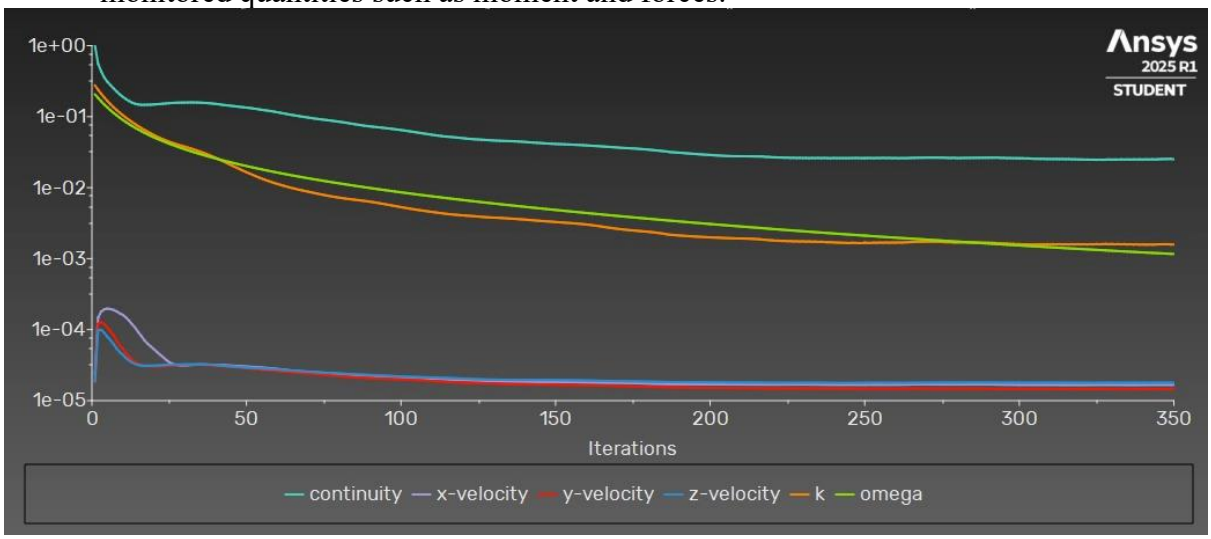
- Turbulence model: k-ω, SST
  - Reason: k-ω is able to capture the turbulence around the blades
- Solver: Pressure-based, steady
  - Reason: since the velocity is low and the flow is incompressible, the pressure-based solver was selected.
- Method: Coupled
  - Reason: coupled method solves the equations simultaneously, which induces more accurate results than other sequential methods.

3.4 Processing (Solution & Convergence)

- In this stage, the numerical solution of the governing equations is carried out after completing the pre-processing steps. The working fluid is defined as air, with constant properties corresponding to standard atmospheric conditions. The air density is set to $\rho = 1.225\ kg/m^3$.
- The rotational speed of the blade is specified as 400 rpm, which is converted into angular velocity for the rotating reference frame according to:
  $\omega = 2\pi(N/60) = 2\pi(400/60) = 41.89\ rad/s$
- A velocity inlet boundary condition is applied with a uniform inlet velocity of 9 m/s. The blade rotation is defined about the X-axis, consistent with the blade orientation in the computational domain. The blade surface is modeled as a no-slip wall, while the outlet is defined as a pressure outlet with zero-gauge pressure.
- From the Report Definitions in ANSYS Fluent, a Force Report is first selected to evaluate the aerodynamic forces acting on the blade. Subsequently, a Moment Report about the X-axis is defined to calculate the aerodynamic moment (torque). The obtained moment value is:
  $M = 19.6\ N \cdot m$
- The mechanical power extracted by the blade is then calculated using:
  $P = \omega \times M$
- The power coefficient (Cp) is evaluated using the standard relation:
  $Cp = P / (\frac{1}{2} \rho A V^3)$

where ρ is the air density, A is the swept area of the rotor, and V is the inlet wind speed.

- Convergence is verified through the reduction of residuals and the stabilization of monitored quantities such as moment and forces.

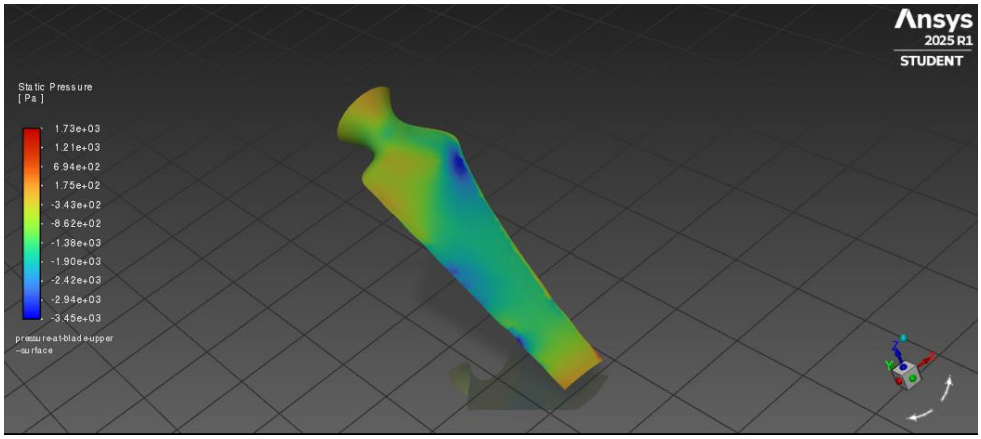


Residual convergence

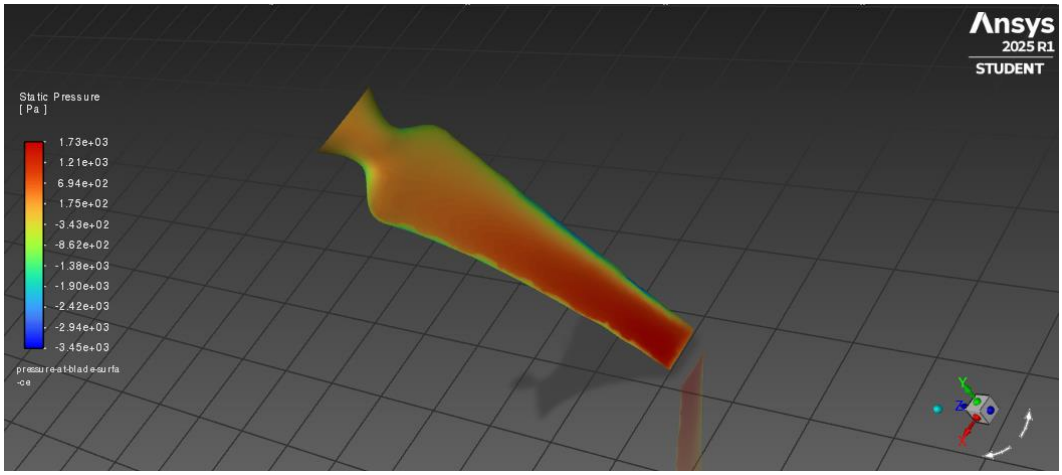
4. Aerodynamic Results (Post-Processing)

4.1 Pressure Contours

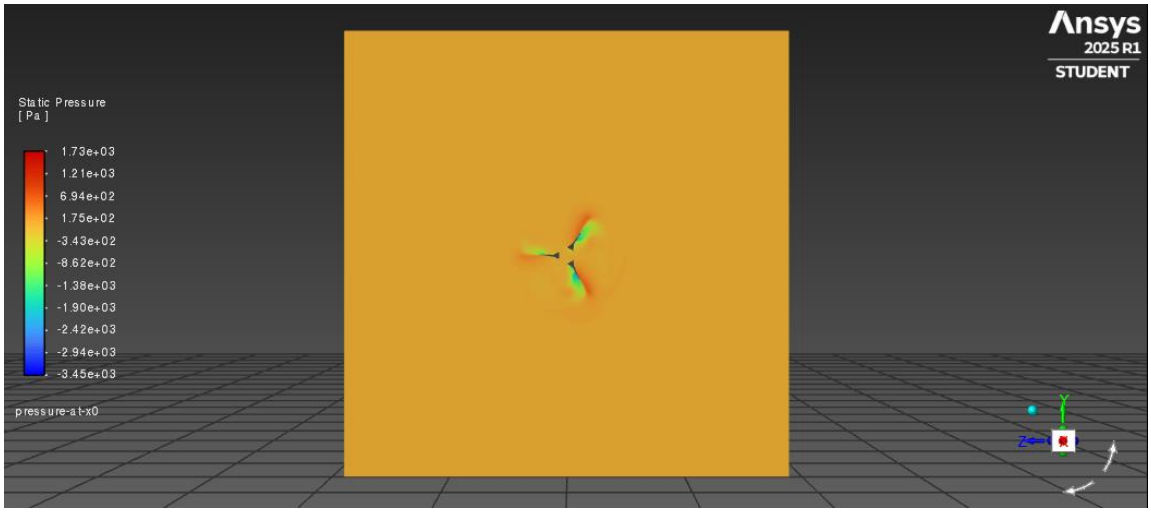
Pressure contours are analyzed to investigate the pressure distribution on the blade and within the flow field. Static pressure contours are extracted as follows:



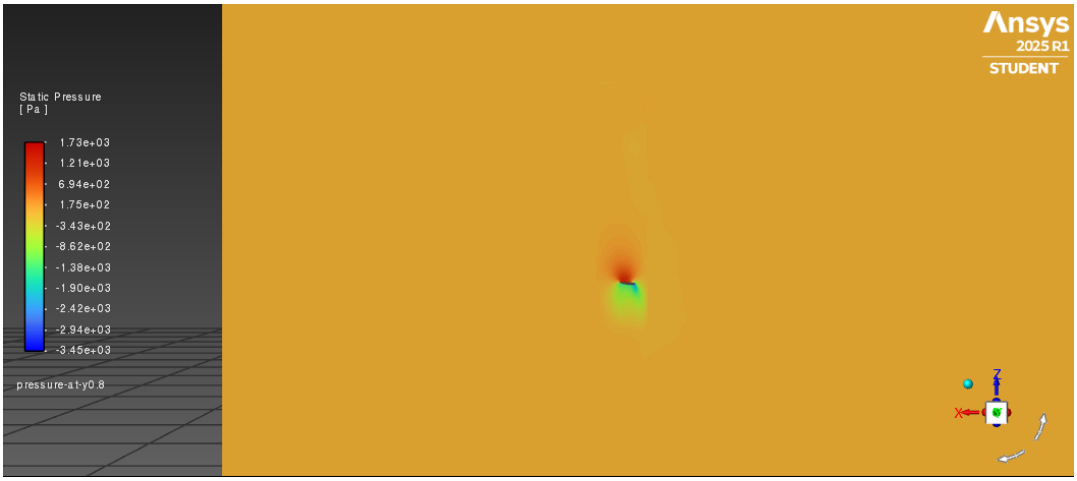
Pressure contours at blade upper surface



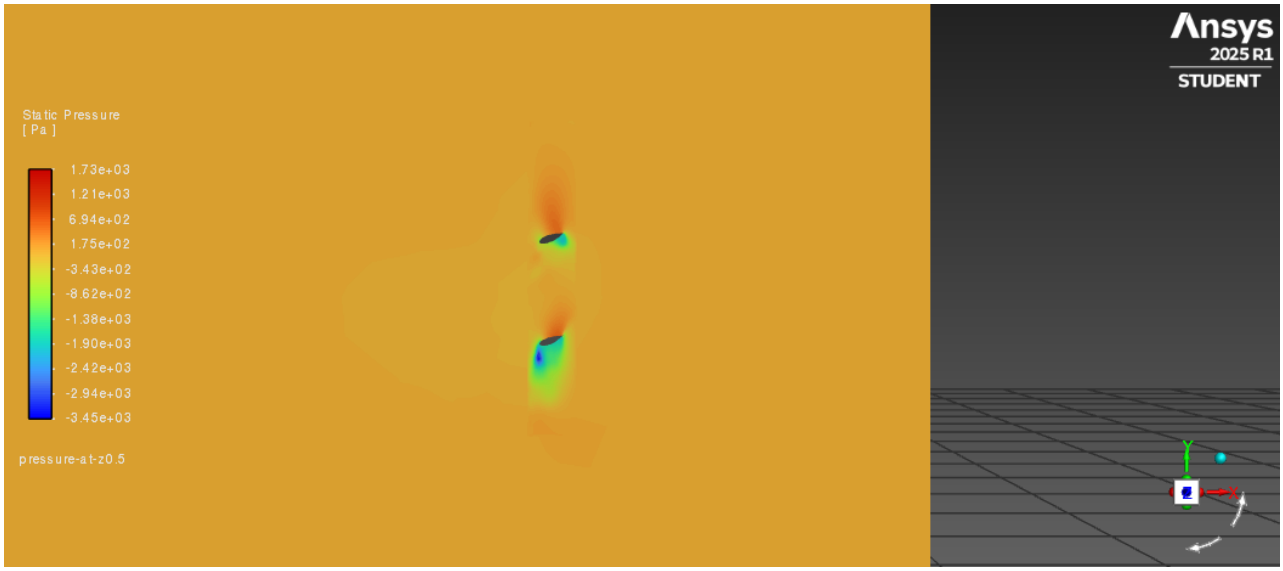
Pressure contours at blade lower surface



Pressure contours at x=0



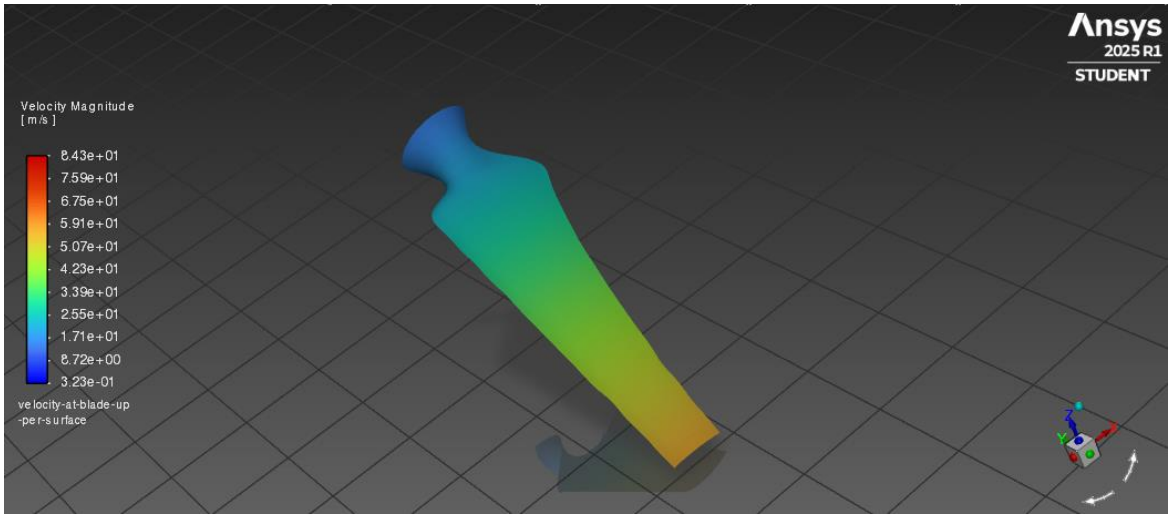
Pressure contours at y=0.8



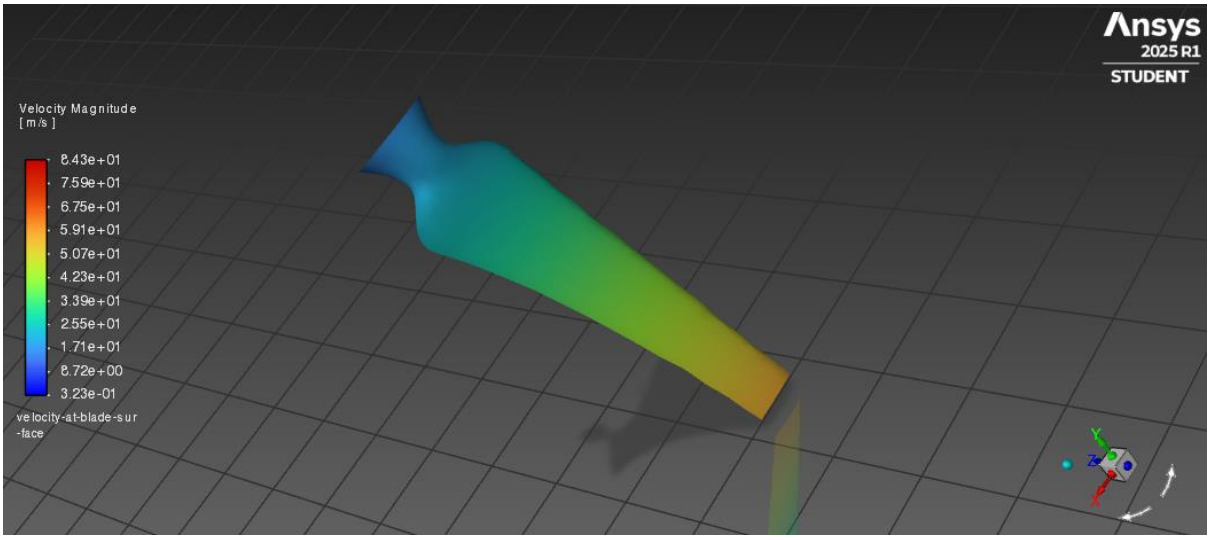
Pressure contours at z=0.5

4.2 Velocity Contours

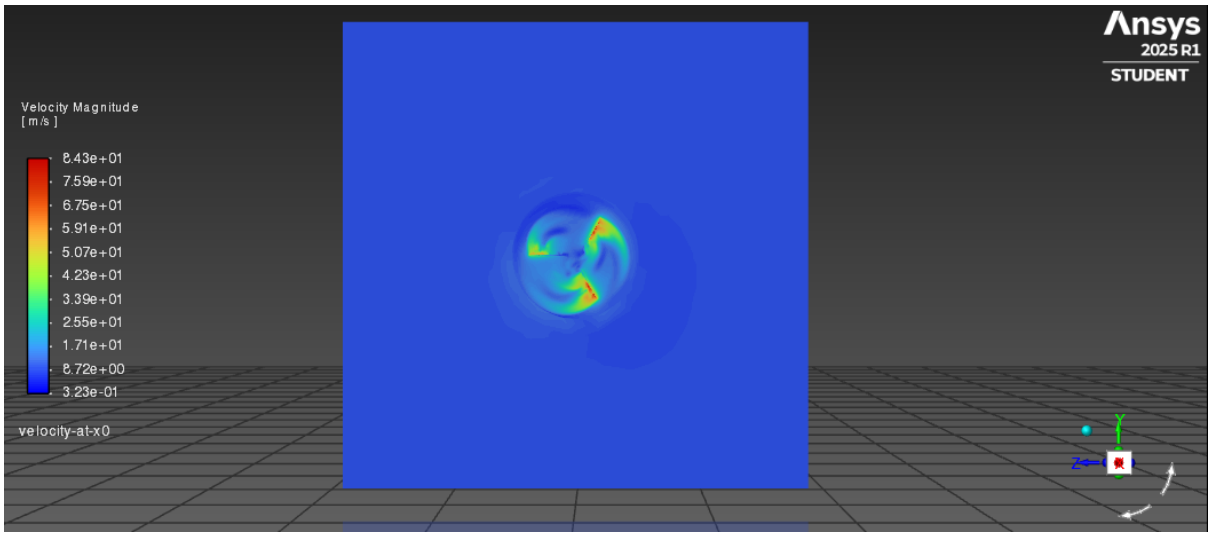
Velocity contours are extracted to analyze the flow behavior around the rotating blade. The velocity magnitude contours are extracted as follows



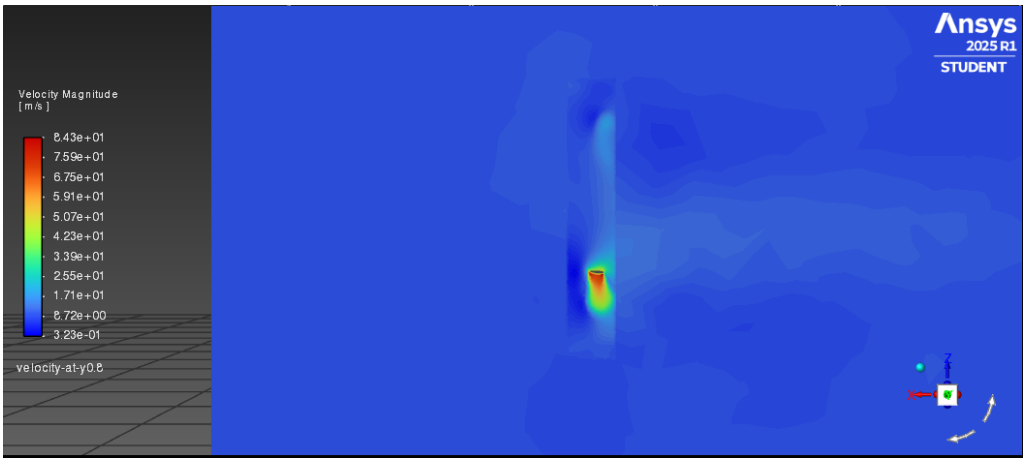
Velocity contours at blade upper surface



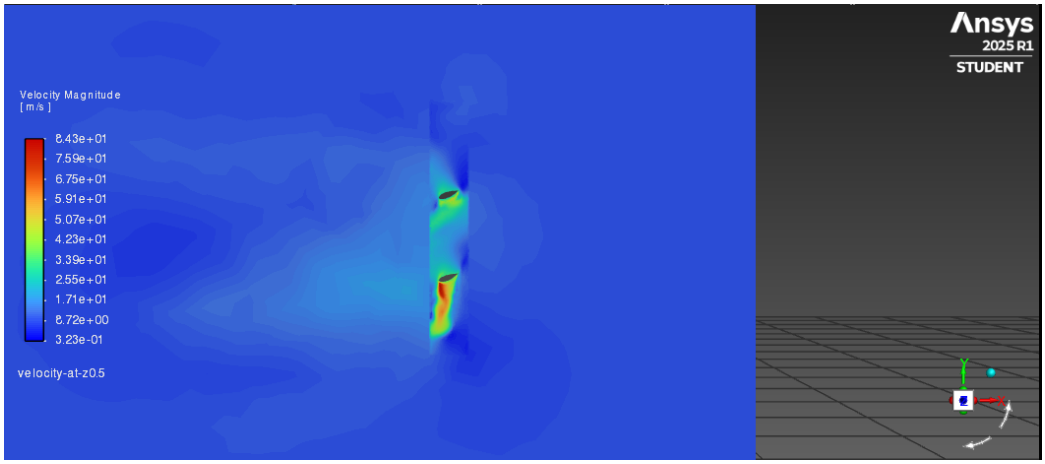
Velocity contours at blade lower surface



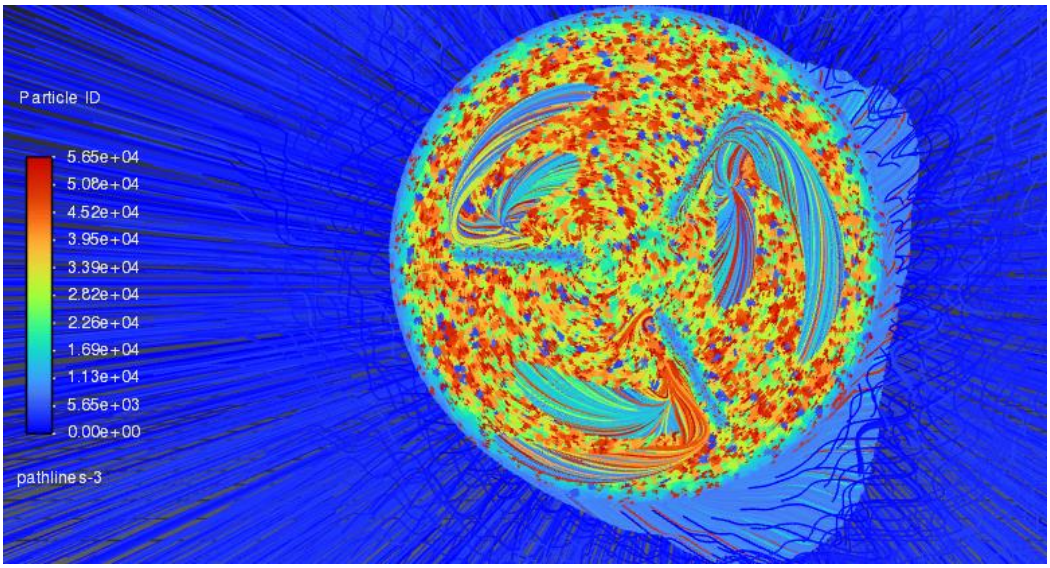
velocity contours at x=0



velocity contours at y=0.8



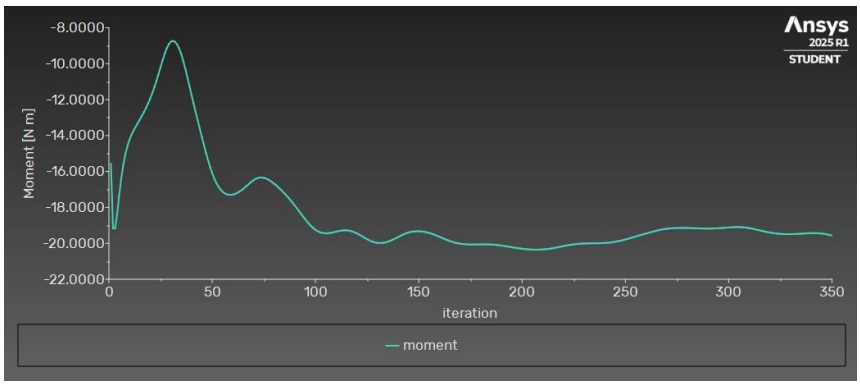
velocity contours at z=0.5



Pathlines colored by velocity magnitude

4.3 Torque, Power, and Power Coefficient

- Torque (from Fluent): T = [19.6] N·m
- Angular speed: $\omega = 2\pi(400/60)$rad/s
- Power: $P = T\omega$= 821 W
- Power coefficient: $C_p = \frac{P}{\frac{1}{2}\rho A V^3}$ = 0.4 (Below Betz's Limit of 0.593)



Moment diagram

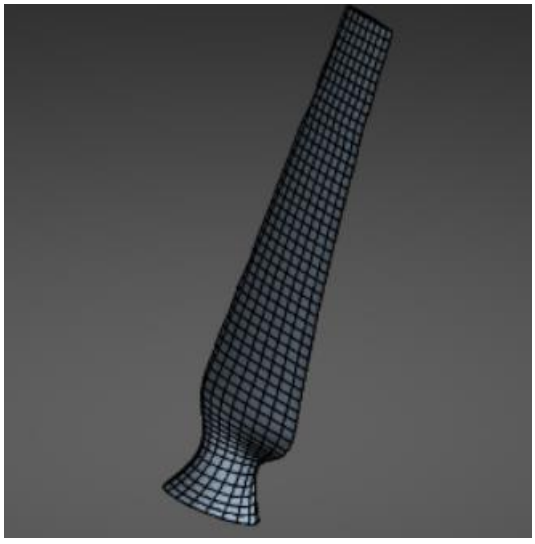
5. Structural Analysis

5.1 Load Transfer and Material

- **Pressure mapping from CFD to Structural**
- **Material: Wood**

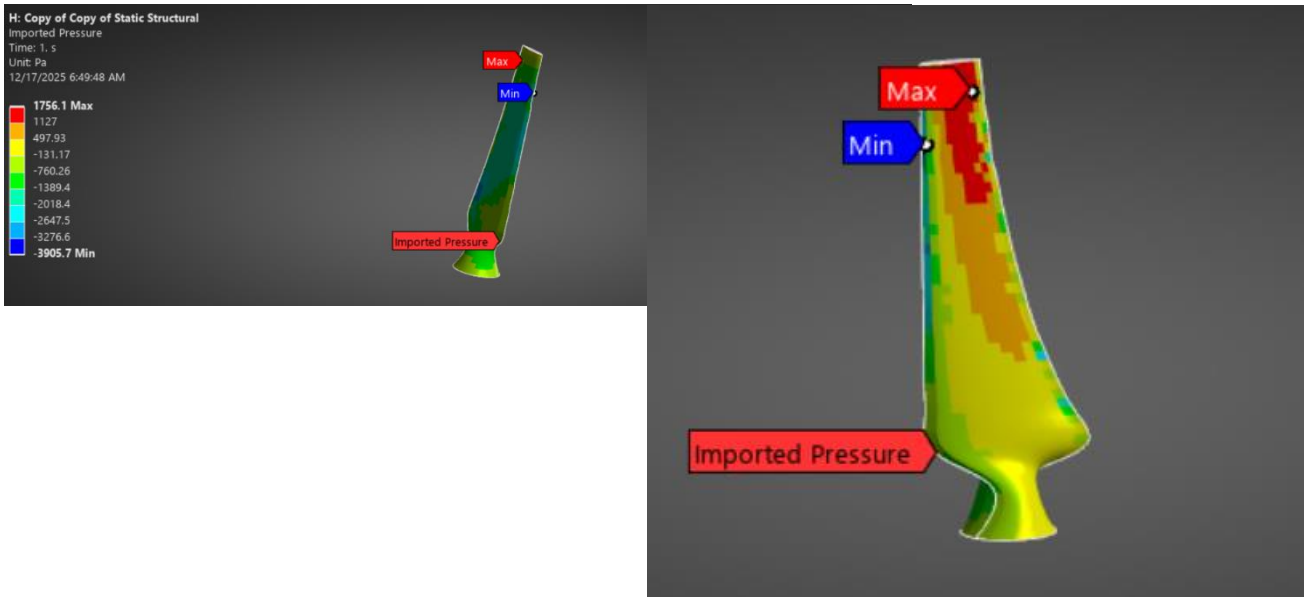
Properties	Value	Unit
Density	553.6	kg/m^3
Isotropic elasticity		
Derived from	Young's modulus	
Young's modulus	89,632	MPa
Poisson's ratio	0.3	
Bulk modulus	7.4693E + 10	Pa
Shear modulus	3.4474E + 10	Pa
Tensile ultimate strength	55.158	MPa
Compressive ultimate strength	31.026	MPa

- **Root constraint:** fixed support at the blade root
- **Meshing:**
  - Mesh type: Prime Mesh
  - Element Size: 0.02m

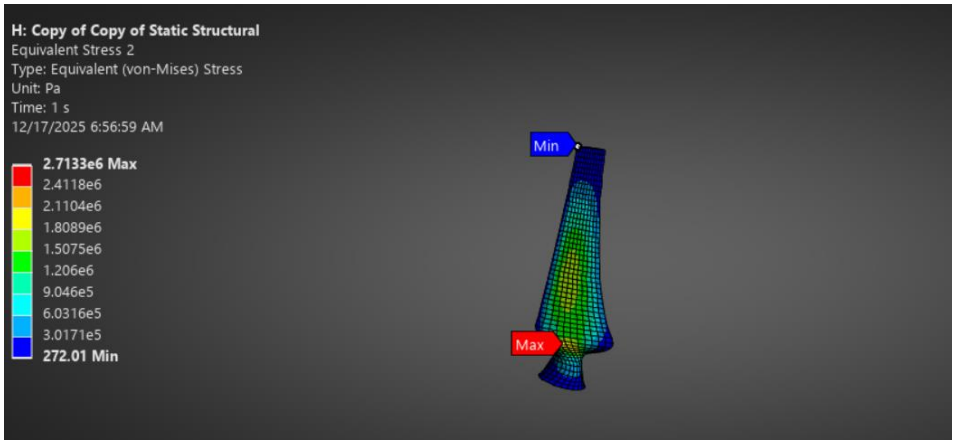


5.2 Stress and Deformation Results

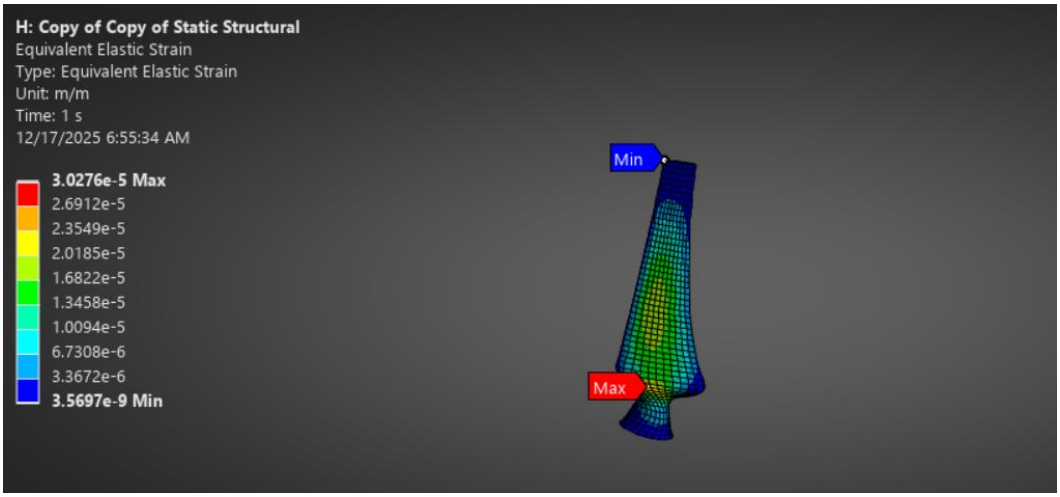
- Pressure on the blade



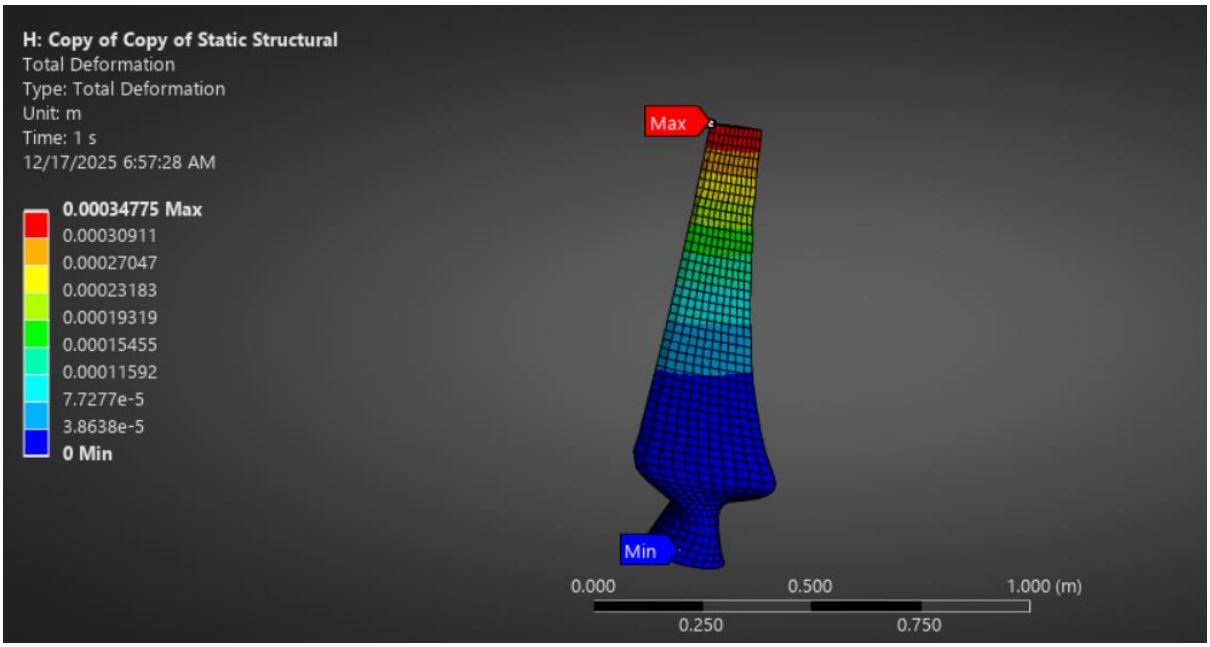
- Equivalent Von Misses Stress



- Equivalent Elastic Strain



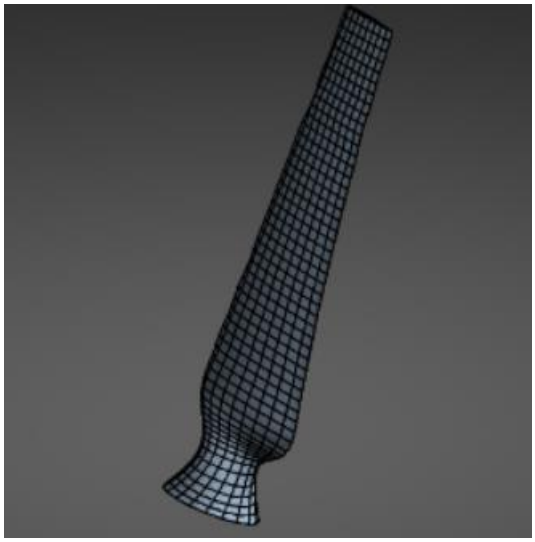
- Total Deformation



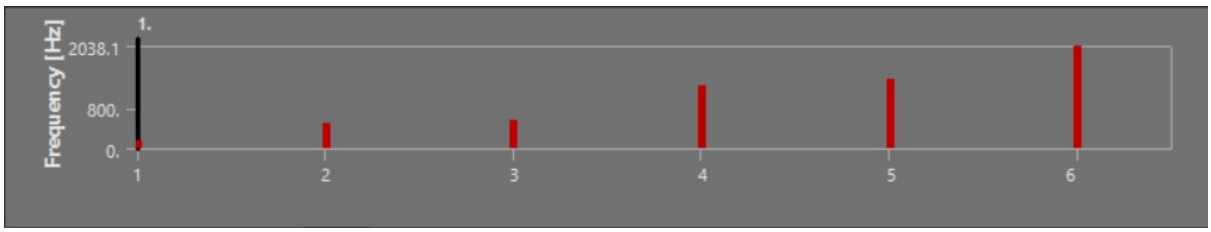
- **Comment:** Since the von-misses stress is below the maximum strength of the material, the blade will not fail

6. Modal Analysis

- **Meshing:**
  - Mesh type: Prime Mesh
  - Element Size: 0.02m

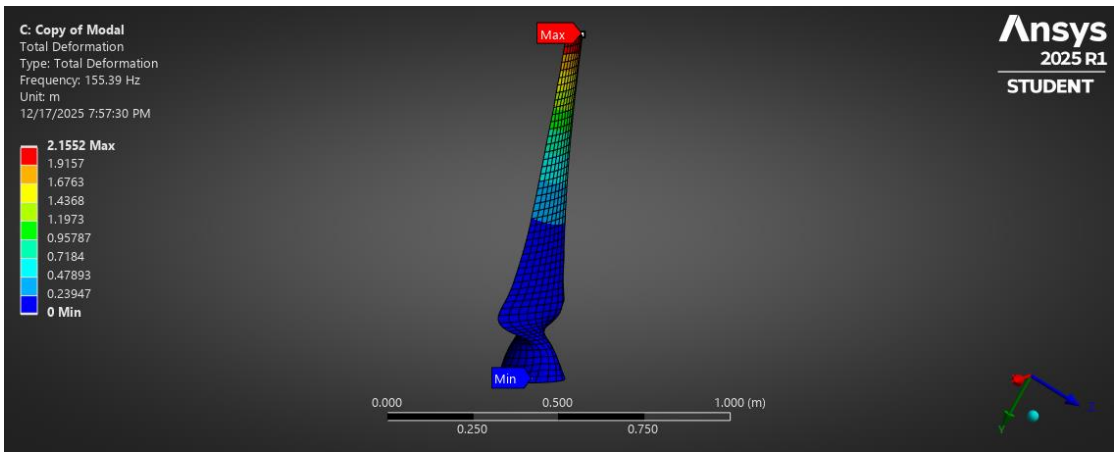


- **Modes and their respective Frequencies**

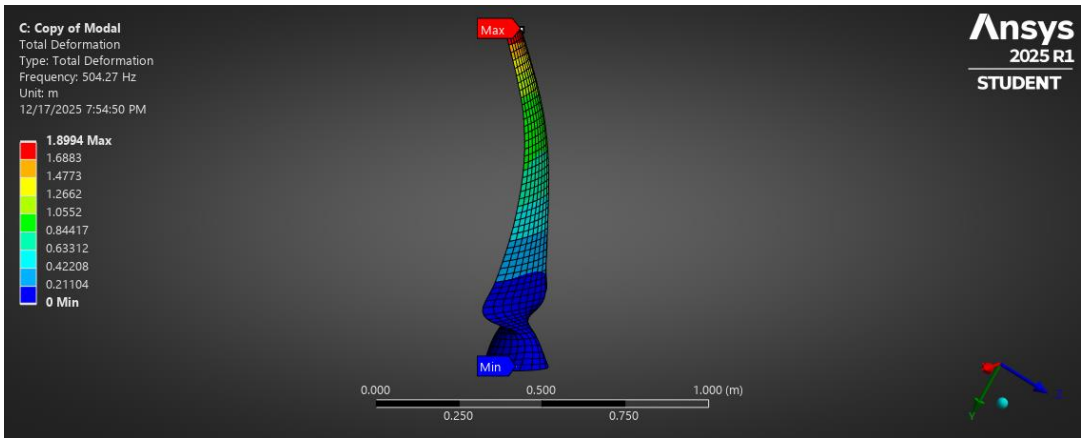


	Mode	Frequency [Hz]
1	1.	155.39
2	2.	504.27
3	3.	558.19
4	4.	1250.5
5	5.	1376.8
6	6.	2038.1

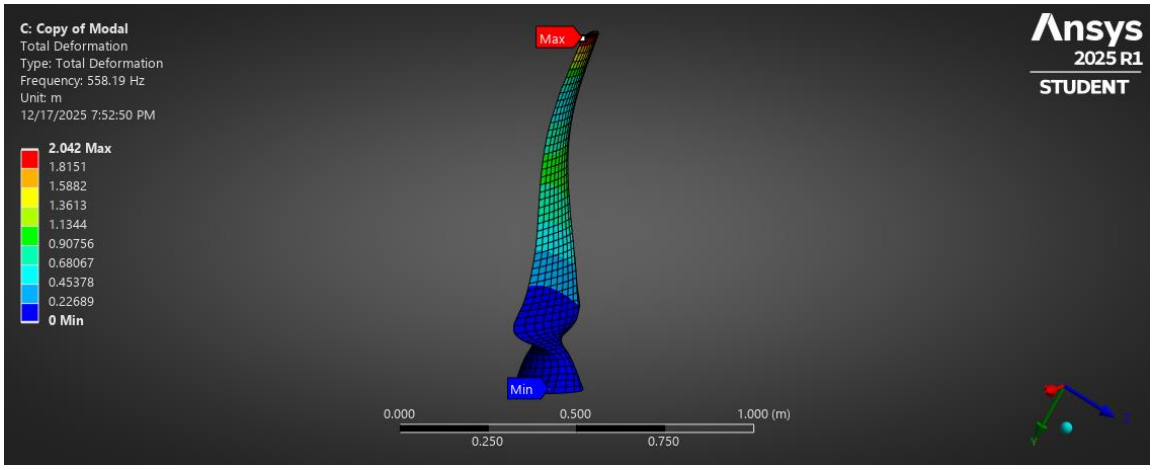
- Total: deformation at mode 1



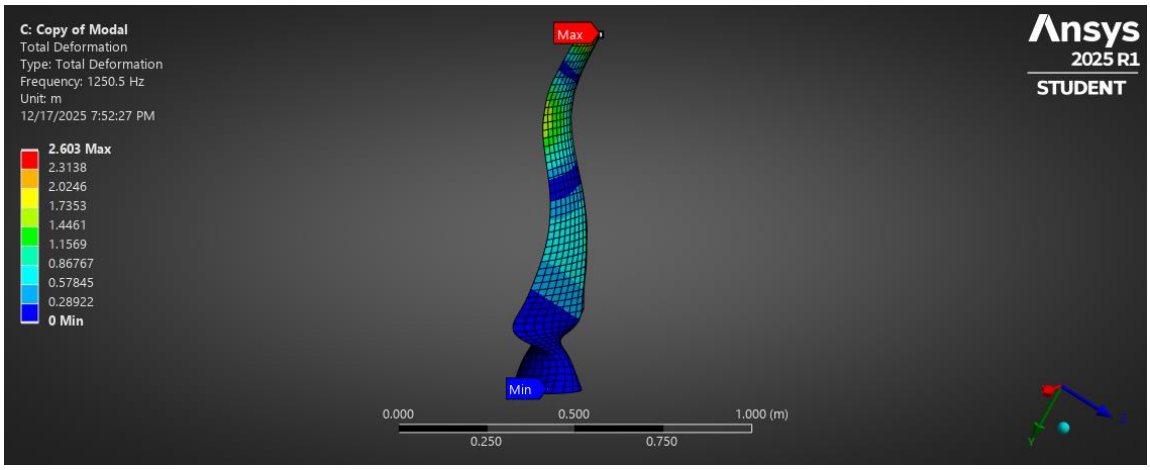
- Total: deformation at mode 2



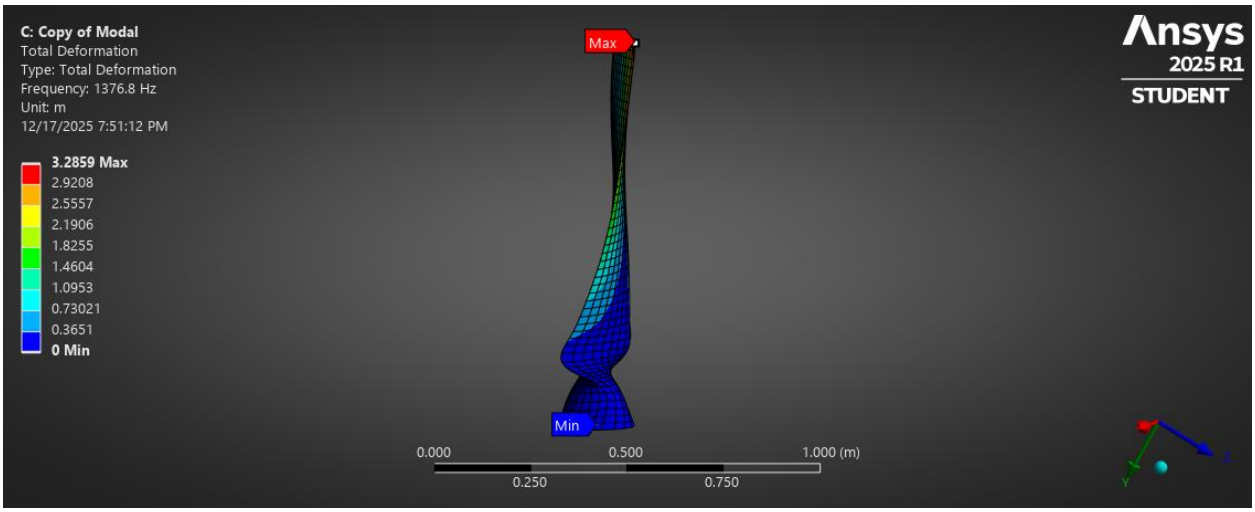
- Total: deformation at mode 3



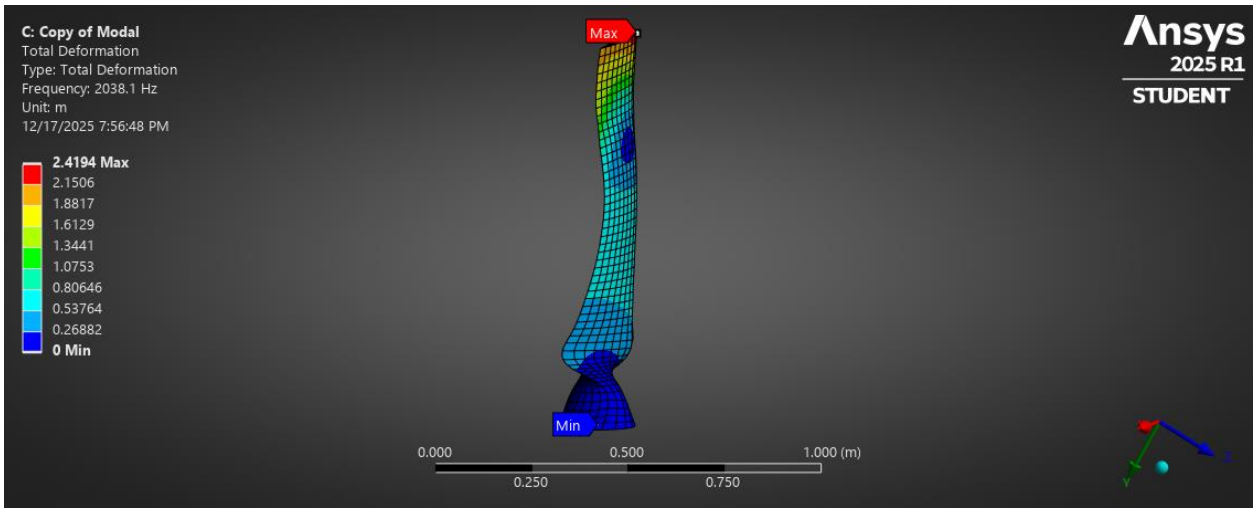
- Total: deformation at mode 4



- Total: deformation at mode 5



- Total: deformation at mode 6



- **Comments:**
  - For the operating speed of 400 rpm, the rotor rotational frequency is 6.67 Hz, with excitation harmonics at 13.3 Hz (2P) and 20 Hz (3P).
  - All extracted natural frequencies are significantly higher than the operational excitation frequencies, with the lowest mode occurring at 155.39 Hz, which is more than 20 times greater than the highest relevant excitation frequency. This large frequency separation ensures that resonance will not occur during normal operation.
  - The modal results confirm that the blade is **dynamically safe** and not susceptible to resonance

7.Conclusion

This project presented a numerical investigation of a horizontal axis wind turbine blade designed using the S834 airfoil. The blade geometry was optimized in QBlade and modeled in SolidWorks to ensure accurate representation of the final design.

A steady three-dimensional CFD analysis was conducted at a wind speed of 9 m/s and a rotational speed of 400 rpm using the k–ω SST turbulence model. The results showed a realistic pressure and velocity distribution, producing a torque of 19.6 N·m, a mechanical power of approximately 821 W, and a power coefficient of 0.4, which is below the Betz limit and indicates efficient aerodynamic performance.

The aerodynamic pressure loads were transferred to a structural model through a one-way fluid–structure interaction approach. Structural analysis demonstrated acceptable stress and deformation levels for the selected wood material. Modal analysis confirmed that the blade's natural frequencies do not coincide with the operating excitation, indicating safe dynamic behavior.

Overall, the results validate the blade design and the adopted numerical methodology, providing a reliable basis for future optimization and advanced wind turbine blade studies.

8.References

[1] Zezatti Flores, M. K., Castro Gómez, L., & Urquiza, G. (2021). *Fluid structure interaction analysis of wind turbine rotor blades considering different temperatures and rotation velocities*. In Computational overview of fluid structure interaction. IntechOpen. https://doi.org/10.5772/intechopen.96495

[2] Wang, L., Quant, R., & Kolios, A. (2016). *Fluid–structure interaction modelling of horizontal-axis wind turbine blades based on CFD and FEA*. Journal of Wind Engineering and Industrial Aerodynamics, 158, 11–25. https://doi.org/10.1016/j.jweia.2016.09.006